4

Supplier HACCP systems

A retailer's perspective

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4.1 Introduction

This chapter looks at the implementation of HACCP systems in the UK from the point of view of retailers. It looks at how major UK retailers have encouraged the implementation of HACCP systems by food manufacturers supplying them with products for their network of supermarkets. The major multiple retailers' interest in HACCP can be traced back to the 1970s when they began to develop their own-brand range of food products, which became commercially highly successful. Supermarkets were acutely aware of their product liability exposure with own-brand products, which were mostly co-produced for them by third-party food processors and packers. Commercially, they needed to be able to demonstrate 'due diligence' in their safety procedures in the event of a criminal or civil prosecution against them, even before this became a statutory requirement. These pressures led to an early interest in HACCP systems.

4.2 Retailers and the development of supplier HACCP systems

Supermarkets quickly built up in-house food technology departments during the 1980s to monitor their own-brand food supply lines and deal reactively with those food safety issues that inevitably arose. They first started to apply formal HACCP techniques in the mid-1980s with the publication by Sainsbury of a supplier guidance manual on HACCP based on the original Pillsbury text of the early 1960s. The other major food retailers soon followed, developing their own approaches to HACCP planning and implementation. Suppliers with more than

one supermarket customer found themselves visited by technologists from each retailer, often with conflicting technical 'advice' on HACCP issues!

Since then the costs of running large in-house technical departments have become a significant competitive disadvantage. In the late 1990s supermarkets began to encourage suppliers to use third-party auditors, approved by the retailers, to audit their safety and quality systems. Concerns were raised about variations in the approach of different third-party auditors. In response to these concerns, the major UK food retailers have, through the British Retail Consortium (BRC), recently agreed a common minimum standard for food safety and quality audits. This standard provides third-party audit companies with a common basis (using HACCP principles) with which to provide a due diligence defence for retailers. UKAS now accredits third-party audit bodies to this BRC standard for retail food supply.

To start with, the major retailers used the HACCP framework as a management troubleshooting tool, because its methodical and logical approach lent itself to the 'reactive' investigation and solution of those food safety incidents that arose on occasions. However, the disruptive costs of product withdrawals and recalls, plus the costs of adverse publicity, soon pushed the supermarkets into developing the use of HACCP systems in a more proactive and preventive way. They began to encourage suppliers to incorporate HACCP principles into their existing quality management systems, i.e. to predict potential food safety issues and build in preventative controls in advance.

Initially, UK retailers found themselves the pioneers in HACCP development, both encouraging their suppliers to adopt HACCP principles and providing the necessary guidance and technical support to allow them to plan and implement HACCP systems. Supermarkets encouraged suppliers in a number of ways, for example by arguing that the successful development of a HACCP system would help a supplier to gain a recognised quality standard such as BS 5750/ISO 9000, at a time when many companies were seeking such accreditation as a source of competitive advantage, though this has now been superseded by the BRC standard. Supermarkets also provided support for suppliers contemplating developing a HACCP system. Sainsbury's first manual for suppliers on HACCP systems, for example, provided a basic template for HACCP planning and implementation, including guidance in such areas as the following:

- selecting a HACCP team and team leader
- constructing detailed process flow charts as a basis for hazard and CCP identification
- procedures for classifying the severity of hazards and isolating CCPs
- procedures for auditing an implemented HACCP plan.

Supermarket staff also provided expertise, for example, on the range of microbiological hazards that suppliers needed to consider, their conditions of growth and methods of control. Initially, the development of HACCP systems was a learning process for both the retailers and the first suppliers implementing

HACCP systems. Some of the problems faced by these suppliers included the following:

- failing to apply the right criteria in hazard and CCP analysis (for instance, by conflating safety and quality issues), resulting in an over-complex and unwieldy initial HACCP design
- unfamiliarity with new systems and responsibilities from line management and staff.

However, supermarkets were able to benefit from their unique position in working with a network of suppliers, able to compare differing approaches to and experiences of HACCP implementation. This experience was used both to strengthen the expertise on which suppliers could draw and develop retailer skills in auditing HACCP systems and making suggestions for improvement. Supermarkets also embarked on a major HACCP training programme for their technologists in the 1990s to consolidate their in-house expertise. Over time, as acceptance and implementation of HACCP principles have become more widespread, supermarkets have increasingly been able to make HACCP implementation a precondition for supplier selection. The technical role of the major retailers has moved towards the administration of a framework of third-party auditing of established supplier HACCP systems.

4.3 Assessing the effectiveness of supplier HACCP systems

The retailers' experience of HACCP implementation by their suppliers has shown that the most successful have been those with a number of common characteristics:

- well-designed and managed prerequisite systems
- existing quality systems, often with certification to a standard such as those within the ISO 9000 series
- a management culture focused on food safety and continuous improvement.

Companies with these characteristics have the right foundation on which to develop a HACCP system. This includes a basic framework of process monitoring and documentation, which can provide a useful starting point for HACCP planning. More importantly, such companies have the enthusiasm and organisational skills to plan and implement a well thought-out HACCP system. It is perhaps not surprising that supermarkets have developed a policy of selecting suppliers from companies with these characteristics, and that, as a result, HACCP implementation has proved a generally rewarding experience for both retailers and suppliers.

There are a number of ways in which supermarkets assess the effectiveness of a supplier's HACCP system. These include:

- regular supplier audits

- monitoring of adverse customer complaints trends, followed by audit visits to isolate and resolve the problems underlying the trend.

The second of these methods is discussed in more detail in the following section which looks at the ways that retailers conduct trend and cluster analysis, based on customer complaint data. Other indicators include:

- variable product quality noticed in routine monitoring
- media reports.

Routine audits assess the effectiveness of HACCP systems in a number of ways. An auditor would first review the documented HACCP plan. An effective assessment of the quality of the plan requires knowledge of the following:

- the product and production processes
- the typical hazards associated with the relevant raw materials and production processes
- the range of controls that should be in place to monitor key processes
- the CCPs and minimum GMP requirements that should be included in the HACCP plan.

Because they have dedicated food safety expertise within their food technology departments, and the privileged experience of working with a range of suppliers on HACCP planning, supermarket staff are able to bring considerable expertise to this initial inspection of a HACCP plan, benchmarking it against other plans and international best practice. One measure of effectiveness is to check whether the plan is under continuous review, for example by looking for revision numbers and dates. This check can show whether the HACCP plan is a 'live' system, responding to changing circumstances, and being effectively implemented and 'owned' by plant management. Another useful indicator of a 'live' HACCP system is the design of CCP monitoring systems and documentation. Clearly set-out procedures for measurement and recording of CCP data, and clear guidance on remedial action in the case of deviation from permitted values, show the care with which the HACCP plan has been put together and is likely to be understood and implemented by line operatives. Review of the HACCP plan would help to determine the audit schedule agreed with the supplier and any special areas of attention in an audit visit to the plant. Modern electronic process control systems can now provide fully integrated HACCP control with corrective actions highlighted at operator interfaces for all CCPs; with the added benefits of reduced paper with improved audit facility.

In the audit visit itself, there are a number of ways of assessing the effectiveness of the implementation of a HACCP plan. These include:

- the use of scoring systems, based on the audit schedule, to provide a more systematic assessment for feedback of overall performance and particular areas for improvement
- questioning department managers', line supervisors' and operatives' understanding about which CCPs they have responsibility for, how frequently they

are monitored and why, and how they deal with any deviations from the permitted values
- sample inspection of monitoring records for selected CCPs to check that measurements are being carried out in the manner set out, and any deviations acted upon.

Speaking to production line staff is particularly important in assessing how effectively a HACCP plan has been implemented and is 'owned' by the relevant staff in the organisation.

Section 4.5 considers a number of common weaknesses in HACCP planning and implementation. These reflect some of the areas of improvement that experienced HACCP exponents have identified in their work with suppliers on the development of effective HACCP systems.

4.4 Gauging the success of HACCP systems: customer complaint data analysis

Supermarkets are in a unique position to analyse how well their suppliers' HACCP systems work. This is because, as the consumer's first port of call, supermarkets are in the front line to capture customer complaint data and analyse it statistically using computer software.

Two systems of customer complaint analysis have been developed by the leading supermarkets:

1. *Trend analysis*, where quality defects are correlated with point of production and used to target quality improvement initiatives or to delete those suppliers who were unresponsive.
2. *Cluster analysis*, where serious customer complaints alleging food poisoning or product contamination are collated as 'clusters' which are analysed for statistically significant correlations. Where the intrinsic risks of the clustered product and the customer complaint symptoms combine in a manner that is capable of professional interpretation as possible food poisoning, an immediate product withdrawal would be actioned and a HACCP-based on-site investigation initiated.

Trend analysis of customer complaint data has long been established in principle, even if in practice the application of statistical techniques had not been uniformly adopted by all supermarkets. Trend analysis allows the identification of the principal sources of product quality problems, and more importantly of the specific causes responsible for the majority of such complaints. In this as in many areas, the 80/20 rule is found to apply, i.e. 80% of problems arise from 20% of the supplier base. Where significant quality problems persist, suppliers may be de-listed by the retailer in question.

Cluster analysis is a more sophisticated concept whereby serious, alleged food poisoning complaints are rapidly screened over a specific time period to

establish whether an unexpected number of such complaints in that time period, or cluster, has occurred. This can be established in advance of any trend in such complaints, allowing for pre-emptive action. It relies upon a professional understanding of the normally arising incidence of complaints, an appreciation of the inherent hazards in any product category and information from customer reports of symptoms of the alleged food poisoning.

Today, all the leading supermarkets have adopted a HACCP-based approach to food safety management where trend and cluster data are key inputs for prioritising the investigative and corrective actions of a more focused team of food technologists either in-house or third-party consultants. Supermarkets have become highly adept at managing crises associated with the common causes of breakdowns in HACCP or other food safety control systems.

4.5 Common weaknesses in HACCP systems

Experience of monitoring and auditing HACCP systems suggests that there are three main areas of weakness:

1. The design of the HACCP plan.
2. Failure to maintain the HACCP system.
3. Very occasionally, management neglect of safety as a priority.

Some examples of these sorts of weakness are discussed below.

4.5.1 Design weaknesses: infant food

In the mid-1990s, the CDSC (Communicable Disease Surveillance Centre), a government agency set up by the Department of Health (DoH) to monitor human disease in the UK population, noted an increase (i.e. a cluster) in the number of an isolate of *Salmonella* food poisoning in very young children. A total of 16 cases of this *Salmonella* isolate had been reported for the first six months of a particular year compared with 12 and 7 cases for the whole of the two previous years. An investigation was begun on the basis that the statistical significance of the data was beginning to indicate a serious incident.

By the end of the investigation, the persistence of CDSC and the DoH at the early stage of this incident, when the statistical evidence was weakest, would be totally vindicated. At the start of the incident, however, two cases dated to two or more months previously, and of the 14 remaining, only ten had been subject to case interview. Of these ten, only six were reported as having consumed Brand X infant food, while five were reported as having consumed Brand Y. While these data appeared slim evidence of causal association, the CDSC statisticians also knew that of their 40 planned control interviews (that is interviews of similar families in the affected areas who had not reported with *Salmonella* food poisoning), the first 16 had not consumed Brand X infant food.

Such statistics, which can be typical of this type of food poisoning incident at the early stages, were daunting to all but the CDSC statisticians. Within the week that the investigation had begun, the DoH decided to contact Brand X to present the data and the emerging suspicions of a causal link. Brand X, sensibly, decided to withdraw all the product from sale immediately on the precautionary principle. Following technical debate well into the late evening, the decision to have a public recall was made that same evening and announced on the following day, and an immediate factory investigation was instigated.

A significant proportion of withdrawn product samples were subsequently found to be *Salmonella* positive, though all at a contamination level significantly below the previously accepted infective dose. The CCP was that the product was designed and marketed as an infant food, which implied a target consumer group with a greater susceptibility to *Salmonella* infection, and therefore a 'lower than average' infective dose. The possibility of an elevated susceptibility of infants to lower than average *Salmonella* contamination levels had not been adequately considered in the product formulation or process specification, because it had not been adequately considered in the initial HACCP study of the product, or identified as a CCP.

4.5.2 Design weaknesses: smoked salmon

Traditionally, smoked salmon was effectively preserved for ambient storage with over 15% total salt (on water content) together with heavy smoking involving up to 30% dehydration weight loss. Such a product was shelf stable at ambient temperature, and suitable for postal distribution. Such heavily salted, smoked and dried traditional products have long ceased to be organoleptically acceptable to today's consumer. Smoked salmon today is only very slightly salted and dried (3.5% salt in water), and effectively only flavoured with the smoking process. As a result, smoked salmon today relies upon a controlled distribution through the refrigerated chill chain for its microbiological stability and safety, within a given shelf-life.

When traditional preservation techniques were phased out, it became clear, from a number of customer complaints and queries to many retailers, that some customers were still sending smoked salmon through the post to friends and relatives. It was evident that some consumers were not aware of the significance of changes in processing and their implications for product handling and storage. Most retailers responded by briefing staff at their stores to warn against this practice, and asking manufacturers to include clear advice in food labelling about the unsuitability of mail delivery for this product, and the need for proper chill chain maintenance.

In hindsight, it was clear that revising HACCP plans to incorporate new processing and preservation techniques had failed to take into account food safety implications further down the supply chain and, in particular, the need to educate consumers to abandon traditional practices such as sending the product through the post. Modern smoked salmon bears little resemblance to the traditional product.

4.5.3 Failure to maintain the HACCP system: ropey milk

The term 'ropey' as applied to milk, beer and bread has come down in the history of the food industry. The term 'ropey' is often used without an awareness of the original food derivation. The 'rope', 'strings' or 'slime' refer to bacterial slime produced by bacilli, that is gelatinous and slimy or sticky in appearance. The problem has been known about for many decades, and has long been solved and relegated to the food science history books. Using cluster analysis an incident was discovered in the early 1990s with some pasteurised milk products where customers were complaining about 'slimy' milk, quickly confirmed as 'ropey' milk.

The first reaction of the management in question when confronted with the cluster analysis data was that, as 'the dairy industry hadn't had a ropey milk problem for 20 years', there must have been some error in identifying the nature of the incident. Three weeks later, and following continuing reports of similar customer complaints of slimy milk, they discovered a grossly contaminated rinse water tank that had been added as a secondary increase in cleaning capacity. The added tank had been configured in such a way that while increasing the total volume of final rinse water available, it also inadvertently created a stagnant volume of water. This stagnant volume of water gradually became contaminated with bacilli, and acted as a contamination source for all the final rinse water used in the dairy.

This incident occurred as a result of a management failure to review the original plant HACCP plan in the aftermath of significant plant changes, i.e. the rinse water tank capacity increase. HACCP plans should always be revised after significant plant changes and additions.

4.5.4 Management neglect: *Salmonella* food poisoning with snack salami

Despite the requirement for 'Best before' date marking, salami is actually a 'Best after' product, but no such labelling designation exists. The reason is that salami is not, as often incorrectly described, a raw meat product. It is a product made from raw meat, but the raw meat protein is denatured by the chemical action of curing salts and bacterial acid production. During this curing process bacterial action and bacterial acid production combine with the curing salts, over time, to produce a safe and delicious food.

The curing process involved in the production of salami requires careful attention to temperature, rate of acidity production and maturation time and conditions. Safe salami product, free from pathogens such as *Salmonella*, can be guaranteed by diligent professional attention to the conditions of production. A traditional product like salami can, however, sometimes be produced where the original understanding of the basic craft and science has been lost, and methods of production are continued 'as they have always been' for generations. The product is still safe as long as the process remains unchanged, even if the original craft and scientific understanding has been lost.

However, in the 1980s, when a new management team decided to produce a snack salami of finger thick dimensions, they changed the process with the fateful result of a major *Salmonella* poisoning incident. The process for normal (approx. 3 inch diameter) salami production was faithfully reproduced, but the critical point was missed that the surface area to mass ratio was critically different. The new snack salami dried much quicker. This meant that the water activity fell faster, crucial microbial activity was suppressed sooner and acidity development was incomplete. Under these new conditions, *Salmonella* was protected from the hostile effects of the curing salts and normal acid production from bacterial fermentation. *Salmonella* tended to survive under these conditions. To make matters worse, any snack product is passed through the stomach more quickly than a normal full meal. In the intestines, where less acidic conditions apply, *Salmonella* can survive more easily to infect the host consumer. This incident demonstrated neglect due to lack of understanding of the basic food science. The hazards arising from the change in process conditions should have been identified by a thoroughly revised HACCP plan.

4.5.5 Management neglect: *Salmonella* contamination of dried baby milk

Ultimately all food safety failures can philosophically be attributed to management failure, but there are some food safety failures that can be attributed to management's failure to learn the lessons of food safety history. The importance of highlighting this problem is not to engage in witch-hunting, but to elevate the importance of continuous training and professionalism in the management cadre.

During the early 1980s a UK brand of dried baby milk suddenly suffered a *Salmonella* contamination problem. The CDSC was capable of statistically associating an outbreak of *Salmonella* food poisoning among babies with a particular Brand Z of baby milk powder. Interestingly, at the start there were no actual contaminated product samples as evidence of product contamination. Looking for actual product contamination presented the proverbial 'needle in a haystack' dilemma. All the evidence was statistical association of disease with product consumption patterns. A curious point was that the *Salmonella* outbreak was all caused by a single strain of *Salmonella* rather than a cocktail of *Salmonellae* strains that normal contamination patterns would create.

In the event it was discovered that a hairline crack in the stainless steel lining of the spray drier had allowed a single cell of *Salmonella* to leak into the rockwool insulation lining of the spray drier. There it had enjoyed a degree of protection from the heat of processing and the chemical sanitisers during cleaning, and with the abundant nutrients of the milk product, had multiplied rapidly. During cyclical processing and cleaning, the *Salmonellae* had migrated to and fro across the stainless steel lining of the spray drier, intermittently contaminating the dried milk product. *Salmonella* is more resistant to heat under dry conditions, and some survived to contaminate the dried milk and subsequently infect some children.

The actual incidence of contamination was very low, hence the difficulty of finding actual contaminated product and the practical impossibility of controlling this problem by product sampling alone, but the epidemiological evidence of *Salmonella* poisoning in the baby population was indisputable. All this was relatively easily determined after the event by judicious professional investigation, using HACCP as an investigative tool, but the question remained as to why the circumstances had not been predicted before the event. The simple answer was that the HACCP studies on the process before the incident had been inadequate. If the whole sequence of events in the incident had never occurred before, the HACCP study would have been limited by previous experience and the failure to predict the problem could have been reasonably accepted. This was not the case, however. The precise sequence of events had occurred in Australia some four years previously. The management in question had failed to keep themselves up to date with events in their industry and apply the lessons from similar incidents.

4.6 The future development of HACCP

Retailers have identified a number of ways in which HACCP design and implementation can fail. These include:

- quite commonly, a failure in the design of the HACCP plan, e.g. in identifying risk levels for particular customer groups; or ineffective hazard analysis because of a failure to keep up to date with current research and best practice
- quite commonly, a failure to audit the HACCP plan satisfactorily and particularly the impact of process changes on the hazards originally identified in the plan
- rarely, a failure of managers to understand the basic food science, processes and hazards sufficiently to undertake a proper hazard analysis, which can only be dealt with by a recognition of the need for training with the appropriate commitment and resources to acquire the requisite knowledge and skills
- very rarely, a failure to identify customer safety as the primary management responsibility, which can only be resolved by appropriate training and changes in management culture.

Given the increasing prominence of food safety issues, these last two failures are now rare, although they have by no means been eliminated as competitive pressures in the food industry intensify. Indeed, they may remain significant problems as global sourcing brings new suppliers into the food chain who may be less familiar with food safety issues and HACCP systems. For those more familiar with HACCP systems, the kind of design and verification problems outlined earlier are likely to continue to be applied as the pace of product innovation intensifies and new hazards emerge. Allied to these problems will be the need to keep established HACCP systems ‘fresh’, for example in

keeping staff motivated. A programme of ongoing staff training will be important here.

The following discussion looks at a number of possible improvements to HACCP design and implementation, from ways of improving current HACCP systems to how the scope of HACCP itself can be extended.

4.6.1 Improving HACCP analysis: improved process flow diagram construction

Constructing an accurate process flow diagram is the critical starting point in a HACCP analysis. In any process, one of the most likely points that a food safety hazard will occur is where an unplanned process delay or interruption happens. Such a process delay point is often the location where build-up of microbiological contamination can occur.

A process flow diagram system, which aids the identification of any likely delay points, is significantly more efficient as a management tool for identifying hazard points in the process. The original process flow diagram method of the American Society of Mechanical Engineers (ASME) still has much to recommend it in this respect. As can be seen from Fig. 4.2, the ASME method allows a detailed analysis of a particular process against seven key criteria which help identify potential problem areas such as process delay points.

4.6.2 Extending the scope of HACCP: criminal malicious product contamination

HACCP systems have sometimes been seen as having a primarily microbiological focus, confined to particular products and processes. However, HACCP principles can be more broadly applied to other aspects of food safety, for example malicious product contamination. Incidents of deliberate malicious product contamination are now, regrettably, to be regarded as an established criminal practice and food industry hazard. All food manufacturers must acknowledge this and therefore plan appropriate countermeasures. ‘Appropriate countermeasures’ simply means elevating the protection of product integrity to the same status as all other food safety control measures that comprise normal good manufacturing practice (GMP).

HACCP principles can be employed and specifically focused on this issue. The experience of managing a technical investigation into a criminal malicious product contamination incident that has occurred within the food chain can be employed to develop measures to prevent a contamination problem in the first place.

Every food processing plant is a unique facility that will inevitably have its own unique security problems. But every processing plant can be reviewed logically using HACCP principles, and sensible measures can be implemented to improve security without creating the fortress conditions that would interfere with the proper functioning of the plant. Every criminal who contemplates

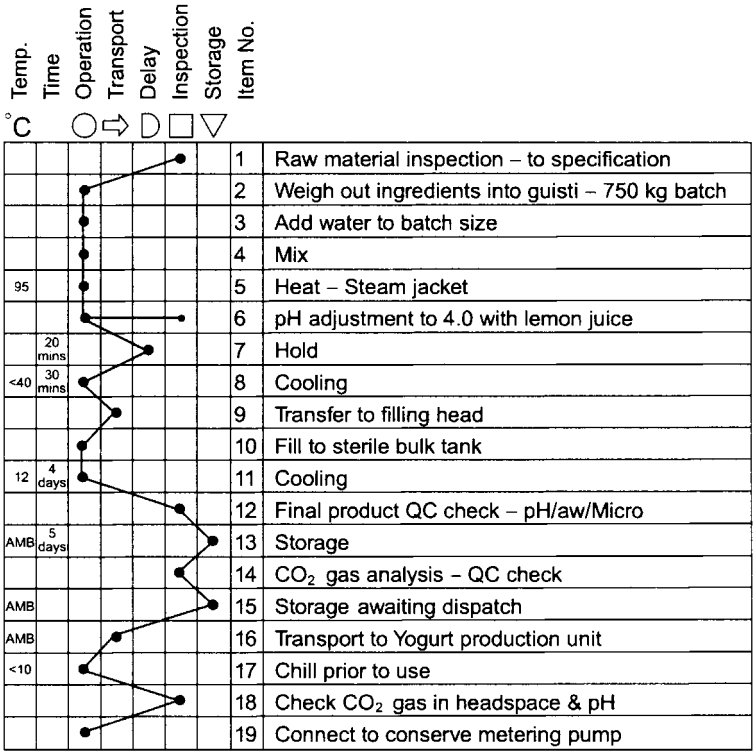


Fig. 4.2 Example of ASME process flow chart (hazelnut yogurt conserve).

deliberate malicious product contamination must have the motive, the means and the opportunity to carry out a crime. HACCP principles can be employed to assist food processors to frustrate potential criminal product contaminators by identifying those with potential motives, restricting the means available and reducing the opportunities to perpetrate their crimes.

Major retailers now issue guidelines to suppliers for dealing with this problem covering such issues as the following:

- staff training in awareness of the problem, identifying potential motives for malicious contamination (e.g. as a result of certain disciplinary actions)
- using a HACCP approach to identify critical points where the product is unavoidably exposed to potential malicious contamination
- security measures (e.g. the use of closed circuit television (CCTV) systems to monitor high risk areas, control of personnel access to high risk areas, colour coding of staff clothing to identify bona fide workers easily in high risk areas, sensors and other instrumentation designed to detect evidence of contamination, tamper-evident packaging)
- crisis management procedures and robust product traceability.

This approach mirrors that used by retailers to deal with malicious contamination in their stores.

4.6.3 Improving the evaluation of HACCP systems: post-launch monitoring of new food products

When new food products are launched, the degree of pre-launch evaluation is limited to panel testing that rarely amounts to more than a few thousand people. More usually only a few hundred people are involved. These numbers bear little statistical significance to the tens of millions of customers that each of the major supermarkets service each week. The customer complaints monitoring that major supermarkets use is also capable of monitoring customer reactions to new products. In the future such existing systems of complaint monitoring could include specific post-launch monitoring of new foods, looking more closely for any emerging food safety problems, for example nutritional issues among more vulnerable sections of the population such as young children or elderly consumers. In fact post-launch monitoring will come to be recognised as an essential element of 'due diligence'. Genetically modified foods could also be monitored by such statistically-based systems. Data from these exercises could be shared with manufacturers of the products concerned, and any general implications for food safety made public so that it can inform future HACCP analysis. In this way HACCP systems can be more effectively evaluated and improved for the future.

4.6.4 Integrated HACCP control systems

The application of HACCP to basic food processing operations calls for the application of specified control procedures at all CCPs, ingredients tracking systems, monitoring and reporting of key factory conditions, cleaning procedures, batch tracking systems and final product traceability. The ease with which these basic requirements can be integrated by modern electronic process control technology has been under appreciated by food manufacturing management, who still currently rely on paper-based control systems.

Of the seven principles of HACCP, four directly address 'process control' aspects. Specifically, these cover:

1. defining the CCPs
2. setting control limits for each CCP
3. monitoring to ensure each CCP is under control
4. taking corrective action when a CCP is out of control.

By definition, a CCP is therefore crying out for the application of automation technology. Where CCPs are controlled automatically, using the full armoury of electronics technologies such as sensing, vision systems, motion control, temperature/process control, operator interfaces, networking, information management and knowledge management, then the production line will benefit

from consistent operation, continuous (fatigueless) monitoring, and the guarantee that these points, procedures or operational steps will be maintained in control within specified limits. And this means safer food production systems, records of non-conformances and corrective actions, full traceability, networked management business process information, vastly reduced paperwork, active operator interfaces for input and instructions, with the final bonus of integrated knowledge management systems based on multi factorial statistical process analysis and diagnostics. The future of HACCP implementation will be in integrated active process control systems using the components of electronic automated process control such as PLCs, touch screen operator interfaces, sensors of many descriptions, PID controllers, distributed controller networks, fieldbus networks, vision systems and RF tagging.

4.7 Conclusions

HACCP is essentially a structured way of thinking about the management of food safety. Its effectiveness depends on the ability of food producers to make the most of the analytical framework it provides and, in particular, the knowledge and skills of HACCP teams in such key areas as hazard analysis and CCP identification. Businesses often fear that the main problem with a HACCP system is the administrative burden they think it will impose. In fact the greater challenge it presents is in analysing the way food products are manufactured and the resulting hazards in a systematic way which some businesses have never previously employed.

In reality, because the quality of HACCP teams varies, the quality of HACCP planning is variable and, in some cases, poor. Retailers auditing supplier HACCP systems also continue to discover within some businesses a lack of understanding of HACCP principles among HACCP teams. The effectiveness of implementation also varies with, in some cases, a lack of ‘ownership’ of the HACCP system by operational staff. Alongside some of the weaknesses identified earlier, common problems include:

- failing to deal with prerequisite systems first as a foundation for HACCP analysis
- over-complex and unmanageable HACCP plans that confuse hazards and quality issues and identify too many control points as ‘critical’
- lack of understanding among CCP monitors of their role, purpose and importance, resulting in poor monitoring and record keeping, and failure to take corrective action.

These problems impose a particular responsibility on the quality of auditing in identifying weaknesses, developing understanding and encouraging initial HACCP systems to continue to improve until they reach a satisfactory standard. Government and industry bodies also have a role to play in spreading understanding of HACCP principles and offering support and advice in such

areas as the training of HACCP leaders and HACCP teams and information on hazards. It will also be important to strengthen the perceived commercial benefits of implementing HACCP systems. One development here is the emergence of new business insurance products which recognise and reward HACCP implementation. Technology will also play a part: improvements in automated process control, for example, will help to make CCP monitoring easier, and developments in real-time, on-line analysis of key hazards will help to validate and improve the design and operation of HACCP systems. Cumulatively, these forces can make the potential of HACCP systems to manage food safety effectively throughout the food industry into a reality that will benefit consumer and industry alike.